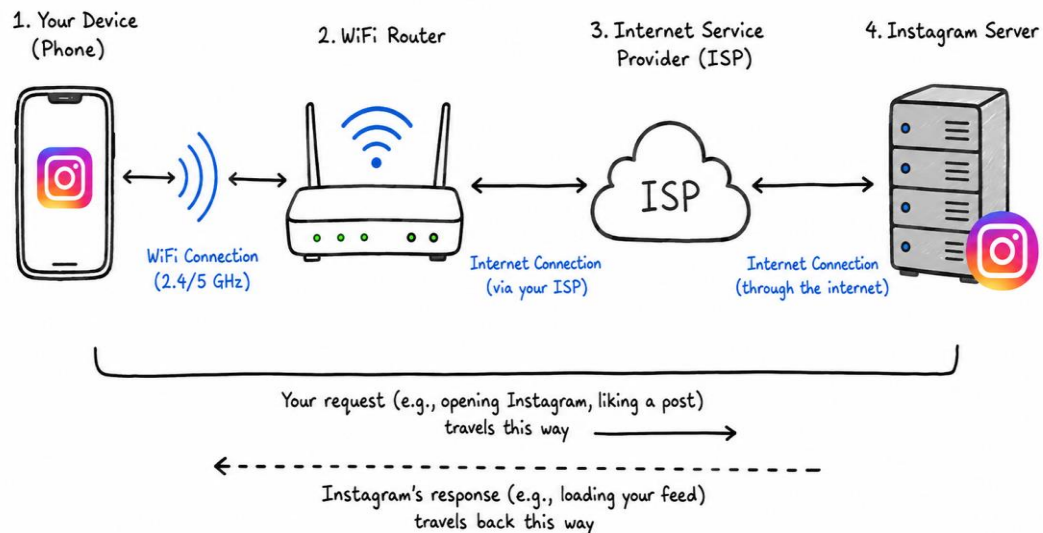


1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, Wi-Fi router, ISP, and Instagram server. Label each part and the connections.



2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, Wi-Fi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

- Here are five common network-connected devices with its network they use:
1. **Smartphone** – WAN (mobile data) and LAN(Wi-Fi)
 - Connects to the internet through a mobile data network (WAN) or a local Wi-Fi network (LAN).
 2. **Laptop** – LAN
 - Connects to a homes, school or office Wi-Fi/Ethernet network to access the internet and shared resources.
 3. **Smart TV** - LAN
 - Connects to a home Wi-Fi or Ethernet network for streaming videos and online apps.
 4. **Wi-Fi Router** – LAN and WAN
 - Creates the local network (LAN) for devices and connects to the internet through ISP's WAN.
 5. **Smartwatch** – PAN (Bluetooth) and sometimes LAN (Wi-Fi)

- Usually connects to a smartphone using Bluetooth (PAN), and some models also connect directly to Wi-Fi (LAN).

3. Explain the difference between the Internet and an intranet using an example from your college or a public Wi-Fi you have used.

- Internet is a worldwide network that connects millions of computers and devices, allowing anyone with an internet connection to access websites, email and online services. It is public and can be accessed from anywhere.
- Intranet is a private network used within an organization, such as a college or company. Only authorised users can access it and it is mainly used to share internal data and resources.
- **Example:-**
- **Internet** – Students use the college Wi-Fi to browse websites, watch educational videos on YouTube or search information on Google.
- **Intranet** – The College’s internal student portal, where students log in to check attendance, download study materials and view exam results. It is only accessible to authorized students and staff.

Internet	Intranet
Public network	Private network
Accessible to all	Accessible to only authorized users
Used for worldwide communication and information	Used for internal communication and resources sharing.
Example : Google, YouTube, Instagram	Example : Student portal or Staff portal

4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

- When I open an app like WhatsApp and type a search, my phone sends the search request over Wi-Fi or mobile data to my internet service provider. The request travels through the internet to WhatsApp’s servers, where the search is processed. The server finds the matching results and sends the response back through the internet. My phone receives the data and app displays the search results on the screen.